We can exploit certain features of the BCH series to obtain more general second order symplectic methods. For Hamiltonians X and Y, consider symmetrizing a composition of exponentials (a Strang splitting), such that

$$\begin{aligned}\exp\left(\frac{t}{2}X\right)\exp\left(tY\right)\exp\left(\frac{t}{2}X\right) &= \exp(tZ_t)\\ &= \exp(t\tilde{Z}_{[1]} + t^2\tilde{Z}_{[2]} + t^3\tilde{Z}_{[3]} + t^4\tilde{Z}_{[4]} + \ldots).\end{aligned}$$

Multiplying by the same expansion using a different time step s, we have

$$\begin{aligned}&\exp\left(\frac{s}{2}X\right)\exp\left(sY\right)\exp\left(\frac{s}{2}X\right)\exp\left(\frac{t}{2}X\right)\exp\left(tY\right)\exp\left(\frac{t}{2}X\right)\\ &\quad= \exp(sZ_s)\exp(tZ_t).\end{aligned} \tag{3.4}$$

But we know that Z_t commutes with Z_s, giving

$$\begin{aligned}\exp\left(sZ_s\right)\exp\left(tZ_t\right) &= \exp\left(sZ_s + tZ_t\right)\\ &= \exp\left((s+t)\tilde{Z}_{[1]} + (s^2+t^2)\tilde{Z}_{[2]} + (s^3+t^3)\tilde{Z}_{[3]}\right.\\ &\quad\left.+(s^4+t^4)\tilde{Z}_{[4]} + \ldots\right).\end{aligned} \tag{3.5}$$

Setting $s = -t$, the left hand side of (3.4) collapses to the identity. Hence the surviving even-powered terms in the expansion on the right hand side of (3.5) must be zero. If the even order terms vanish, then the effect of composing linear operators symmetrically is to keep only odd order terms in the exponent, giving the symmetric BCH formula

$$\exp\left(\frac{t}{2}X\right)\exp\left(tY\right)\exp\left(\frac{t}{2}X\right) = \exp(t\left(X+Y\right) + t^3\tilde{Z}_{[3]} + \ldots), \tag{3.6}$$

where

$$\tilde{Z}_{[3]} = \frac{1}{12}[Y,[Y,X]] - \frac{1}{24}[X,[X,Y]]. \tag{3.7}$$

More complicated symmetric second-order schemes can be devised by proceeding almost arbitrarily and maintaining a symmetric composition, but it is not found that alternative approaches improve on the two Verlet schemes, at least for molecular applications. The Verlet methods are seen as the gold-standard for molecular dynamics computations: both require only one evaluation of $\nabla U(\boldsymbol{q})$ per iteration (where the velocity Verlet scheme can reuse $\nabla U(\boldsymbol{Q})$ for the next iteration), and offer a second-order symplectic evolution.